SAMPLE PAPER DOUBTS

1. Information

From an information theory perspective, "information" is not about meaning but about **uncertainty or surprise**. The occurrence of a highly probable event conveys little information (since it was expected), while the occurrence of a highly improbable event conveys a lot of information (since it reduces a great deal of uncertainty).

The **self-information** of an event x with probability $p(x)$ is defined as:

$$I(x) = \log_2\left(\frac{1}{p(x)}\right) = -\log_2 p(x)$$

- **Properties:**
  - Information is a decreasing function of probability: $p(x) \downarrow \Rightarrow I(x) \uparrow$.
  - $I(x) \geq 0$. Information is non-negative.
  - The unit of information is the **bit** if the log is base 2. It is the **nat** if the natural log is used .

2. Entropy

Entropy, denoted $H(X)$, is the **average or expected amount of information** produced by a random source X. It quantifies the average uncertainty associated with the random variable's possible outcomes.

For a discrete random variable X with probability mass function $p(x)$:

$$H(X) = -\sum_{x \in \mathcal{X}} p(x) \log_2 p(x) \text{bits}$$

- **Properties:**
  - $H(X) \geq 0$. Entropy is non-negative .
  - $H(X) \leq \log_2 |\mathcal{X}|$, where $|\mathcal{X}|$ is the number of outcomes. Entropy is maximized when all outcomes are equally likely.
  - **Joint Entropy**: $H(X,Y) = -\sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p(x,y) \log_2 p(x,y)$.
  - **Conditional Entropy**: $H(Y \mid X) = \sum_{x \in \mathcal{X}} p(x) H(Y \mid X = x) = -\sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p(x,y) \log_2 p(y \mid x)$.
  - **Chain Rule**: $H(X,Y) = H(X) + H(Y \mid X) = H(Y) + H(X \mid Y)$.

3. Mutual Information

Mutual information, $I(X;Y)$, is a measure of the **amount of information that one random variable contains about another**. It quantifies the reduction in uncertainty of one variable due to the knowledge of the other.

It is defined as:

$$I(X;Y) = \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p(x,y) \log_2 \frac{p(x,y)}{p(x)p(y)}$$

- **Properties:**
  - **Non-negativity**: $I(X;Y) \geq 0$. It is zero if and only if X and Y are independent .
  - **Symmetry**: $I(X;Y) = I(Y;X)$. Information about X from Y is the same as information about Y from X .
  - **Relation to Entropy**: $I(X;Y) = H(X) - H(X \mid Y) = H(Y) - H(Y \mid X) = H(X) + H(Y) - H(X,Y)$.
  - **Self-Information**: $I(X;X) = H(X)$. The mutual information of a variable with itself is its entropy .
  - **Chain Rule**: $I(X;(Y,Z)) = I(X;Y) + I(X;Z \mid Y)$.

4. Source Coding Theorem (Noiseless Coding Theorem)

This theorem, also known as Shannon's first theorem, establishes the fundamental limits on **data compression**. It states that a source with entropy H can be encoded with an average code length arbitrarily close to H bits per symbol, but not less than H. Conversely, any coding scheme that attempts to compress the data to an average length of less than H bits per symbol will lose information (i.e., will not be uniquely decodable) .

5. Kraft's Inequality

Kraft's inequality gives a **necessary and sufficient condition for the existence of a prefix-free code** (also known as an instantaneous code). For a code with D symbols in its alphabet and codeword lengths $l_1, l_2, \ldots, l_m$, a prefix code exists if and only if:

$$\sum_{i=1}^{m} D^{-l_i} \leq 1$$

This inequality is fundamental in proving the optimality of coding schemes like Huffman coding .

6. Linear Block Code

In coding theory, a linear block code is an **error-correcting code** where any linear combination (over a finite field) of codewords is also a codeword. It takes a block of k information bits and encodes it into a block of n coded bits (hence, an (n, k) code). The encoding is done using a generator matrix $\mathbf{G}$. These codes are valued for their structured properties, which simplify encoding and decoding.

7. Channel Capacity

Channel capacity, C, is the **maximum rate at which information can be reliably transmitted over a communication channel**. Reliable transmission means that the probability of error can be made arbitrarily small. It is defined as the maximum over all possible input distributions $p(x)$ of the mutual information between the channel's input X and output Y:

$$C = \max_{p(x)} I(X;Y)$$

The **Channel Coding Theorem** (Shannon's second theorem) proves that for any rate $R < C$, there exists a code that allows for transmission with an arbitrarily small error probability

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Relationship Between Entropy and Mutual Information

Definition of Mutual Information

Mutual information $I(X;Y)$ measures the amount of information that one random variable contains about another. For discrete random variables X and Y with joint probability mass function $p(x,y)$ and marginals $p(x)$ and $p(y)$, it is defined as:

$$I(X;Y) = \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p(x,y) \log \frac{p(x,y)}{p(x)p(y)}.$$

The logarithm is usually taken base 2, giving units of bits.

Relationship with Entropy

Mutual information can be expressed in terms of entropies. Recall the definitions:

- **Entropy**: $H(X) = -\sum_x p(x) \log\, p(x)$
- **Joint entropy**: $H(X,Y) = -\sum_{x,y} p(x,y) \log\, p(x,y)$
- **Conditional entropy**: $H(Y \mid X) = -\sum_{x,y} p(x,y) \log\, p(y \mid x)$

Using the chain rule $H(X,Y) = H(X) + H(Y \mid X) = H(Y) + H(X \mid Y)$, we derive:

$$\begin{aligned} I(X;Y) &= \sum_{x,y} p(x,y) \log \frac{p(x,y)}{p(x)p(y)} \\ &= \sum_{x,y} p(x,y) \log\, p(x,y) - \sum_{x,y} p(x,y) \log\, p(x) - \sum_{x,y} p(x,y) \log\, p(y) \\ &= -H(X,Y) - \sum_{x} p(x) \log\, p(x) - \sum_{y} p(y) \log\, p(y) \\ &= -H(X,Y) + H(X) + H(Y). \end{aligned}$$

Thus,

$$I(X;Y) = H(X) + H(Y) - H(X,Y).$$

Also, from the chain rule,

$$I(X;Y) = H(X) - H(X \mid Y) = H(Y) - H(Y \mid X).$$

These relationships show that mutual information is the reduction in uncertainty of one variable due to knowledge of the other.

Proof that Mutual Information is Non-Negative

We need to show $I(X;Y) \geq 0$. This follows from the convexity of the logarithm or from Gibbs' inequality.

Proof using Jensen's inequality:

Consider the Kullback-Leibler divergence (relative entropy) between the joint distribution $p(x,y)$ and the product distribution $p(x)p(y)$:

$$D(p(x,y) \parallel p(x)p(y)) = \sum_{x,y} p(x,y) \log \frac{p(x,y)}{p(x)p(y)} = I(X;Y).$$

The divergence is always non-negative, with equality iff $p(x,y) = p(x)p(y)$ (i.e., independence). This is a standard result: for any two probability distributions P and Q, $D(P \parallel Q) \geq 0$ by Gibbs' inequality, which follows from the log sum inequality or from Jensen's inequality applied to the convex function $-\log\ x$.

Specifically, since $-\log$ is convex, we have:

$$D(P \parallel Q) = \sum P(i) \log \frac{P(i)}{Q(i)} = -\sum P(i) \log \frac{Q(i)}{P(i)} \geq -\log \left(\sum P(i) \frac{Q(i)}{P(i)} \right) = -\log\ 1 = 0.$$

Thus, $I(X;Y) \geq 0$, and equality holds iff X and Y are independent.

Kraft's Inequality

Statement

For a D-ary code (codewords composed of symbols from an alphabet of size D), let the codeword lengths be $l_1, l_2, \ldots, l_m$. A necessary and sufficient condition for the existence of a **prefix-free code** (also called an instantaneous code) with these lengths is

$$\sum_{i=1}^{m} D^{-l_i} \leq 1.$$

Proof

Necessity (If a prefix code exists, then the inequality holds)

Consider a full D-ary tree of depth L, where L is the maximum codeword length. Each codeword corresponds to a leaf at depth l_i in the tree. In a prefix code, no codeword is an ancestor of another, so the subtrees rooted at each codeword are disjoint.

Now, at depth L, the total number of possible nodes is D^L. For a codeword of length l_i, it "blocks" D^{L-l_i} leaves at depth L (all descendants). Since these sets of leaves are disjoint for different codewords, the total number of blocked leaves is $\sum_i D^{L-l_i}$, which cannot exceed the total number of leaves D^L. Hence,

$$\sum_{i=1}^{m} D^{L-l_i} \leq D^L \Rightarrow \sum_{i=1}^{m} D^{-l_i} \leq 1.$$

Sufficiency (If the inequality holds, a prefix code can be constructed)

Given lengths $l_1 \leq l_2 \leq \cdots \leq l_m$ satisfying $\sum D^{-l_i} \leq 1$, we can construct a prefix code by the following greedy algorithm (or using a tree):

1. List the codewords in order of increasing length.
2. For each length l_i, assign a codeword that has not been used as a prefix. This is possible because at each step, the number of available codewords of length l is at least D^l times the remaining "probability mass". A formal proof uses the fact that the inequality ensures we never run out of available nodes.

Alternatively, consider a D-ary tree. We want to assign nodes at depths l_i such that no node is an ancestor of another. The condition $\sum D^{-l_i} \leq 1$ is exactly the condition that the sum of the "probabilities" (or weights) is at most 1, which is necessary and sufficient for the existence of such a tree by the Kraft–McMillan theorem. A constructive method is to start with a full tree and prune it.

Thus, the inequality is both necessary and sufficient for the existence of a prefix code.

SOLUTION OF CHAPTER 1

1.1 Can differential entropy be negative? Explain.

Yes, differential entropy can be negative .
Explanation: Differential entropy is the entropy of a continuous random variable, defined as $h(X) = -\int f(x)\log f(x)dx$. Unlike discrete entropy, which is always non-negative, differential entropy is a measure of dispersion relative to the unit of measurement. If a distribution is very concentrated (e.g., a uniform distribution on an interval of length less than 1), its differential entropy can be negative . For example, if $X \sim U(0,0.5)$, then $h(X) = \log_2(0.5) = -1$ bit. This negativity does not violate any information-theoretic principle but rather highlights that differential entropy is not an absolute measure of information content in the same way discrete entropy is .

1.2 We require an infinite number of bits to represent a continuous random variable precisely. True or false?

True.
Explanation: A continuous random variable can take on an uncountably infinite number of possible values. To represent it *precisely* (with infinite precision) would require an infinite number of bits. In practice, we quantize continuous values to a finite level of precision, which requires a finite number of bits .

1.3 If X and Y are independent, is it true that $H(X,Y) = H(X) + H(Y)$?

Yes, it is true.
Explanation: The joint entropy is defined as $H(X,Y) = H(X) + H(Y \mid X)$. If X and Y are independent, then knowing X provides no information about Y, so $H(Y \mid X) = H(Y)$. Substituting this into the chain rule gives $H(X,Y) = H(X) + H(Y)$.

1.4 Let X, Y, and Z be discrete random variables with $I(X;Y) = 0$ and $I(X;Z) = 0$. Is $I(X;Y,Z) = 0$?

No, not necessarily.
Explanation: $I(X;Y) = 0$ means X and Y are independent. $I(X;Z) = 0$ means X and Z are independent. However, Y and Z together might still contain information about X. For example, if $X = Y \oplus Z$ (XOR of Y and Z), with Y and Z being independent fair coins, then X is independent of Y alone and independent of Z alone. But given both Y and Z, X is completely determined, so $I(X;Y,Z) = H(X) = 1$ bit, which is not zero .

1.5 Does conditioning reduce entropy? Is $H(X \mid Y) \leq H(X)$? Is $h(X \mid Y) \leq h(X)$?

- **For discrete entropy H:** Yes, **conditioning reduces entropy**. Formally, $H(X \mid Y) \leq H(X)$. This is because knowing Y can only reduce or leave unchanged the uncertainty about X. Equality holds if and only if X and Y are independent .
- **For differential entropy h: No, this is not always true.** The inequality $h(X \mid Y) \leq h(X)$ does **not** hold in general for differential entropy .

1.6 When is $H(X \mid Y) = H(X)$?

$H(X \mid Y) = H(X)$ if and only if X **and** Y **are independent**. This is because if Y provides no information about X, then the uncertainty about X given Y is the same as the original uncertainty about X .

1.7 Is the function $\log(x)$ concave or convex?

The function $\log(x)$ (for $x > 0$) is **concave**. This can be verified by noting that its second derivative is negative ($\frac{d^2}{dx^2}\log(x) = -\frac{1}{x^2} < 0$) .

1.8 Is the function $x\log(x)$ concave or convex?

The function $x\log(x)$ (for $x > 0$) is **convex**. Its second derivative is positive ($\frac{d^2}{dx^2}(x\log x) = \frac{1}{x} > 0$). This property is crucial for proving important inequalities like Gibbs' inequality and the non-negativity of relative entropy .

1.9 What is the value of $I(X;X)$?

$I(X;X) = H(X)$. This is the **self-information** or information that a random variable conveys about itself. It is simply the entropy of X .

1.10 Can mutual information be negative?

No. Mutual information is always non-negative ($I(X;Y) \geq 0$). It is zero if and only if X and Y are independent .

1.11 Is {1, 01, 001, 000} a prefix code? What about {00, 01, 10, 11}?

- **{1, 01, 001, 000}: No**, this is **not** a prefix code. The codeword "1" is a prefix of "001" and "000"? Actually, no, "1" is not a prefix of "001" (which starts with 0). Let's re-examine: The set is {**1**, **01**, **001**, **000**}. Is any codeword a prefix of another?
    - Is "1" a prefix of "01"? No, "01" starts with 0.
    - Is "1" a prefix of "001"? No.
    - Is "01" a prefix of "001"? Yes! The codeword "01" is exactly the start of the codeword "001". Therefore, it fails the prefix condition.
- **{00, 01, 10, 11}: Yes**, this **is** a prefix code. All codewords are the same length (2 bits), so no codeword can be a prefix of another .

1.12 Does the addition of independent random variables increase uncertainty?

Generally, **yes, entropy increases**. If X and Y are independent, then $H(X+Y) \geq \max(H(X), H(Y))$. The sum typically has more possible outcomes and is more "spread out" than either variable individually, leading to higher uncertainty. There is a specific inequality that can be proven: $H(X+Y) \geq H(X)$.

1.13 By what factor does one bit reduce the expected distortion?

One bit of information reduces the expected distortion by a factor of $\frac{1}{4}$ (or reduces the squared error by a factor of 4). This is a key result from **rate-distortion theory**. For a Gaussian source, the distortion-rate function is $D(R) = \sigma^2 2^{-2R}$. So, adding one bit ($R \to R+1$) multiplies the distortion by $2^{-2} = \frac{1}{4}$.

1.14 You have 12 identical gold coins of which one is a counterfeit (weighs either more or less than others). You have a beam balance. What is the minimum number of weighs needed to identify the counterfeit coin? How is it related to information theory?

- **Minimum number of weighings: 3 weighings** .
- **Relation to Information Theory:**
  This classic puzzle is directly related to information theory. Each weighing has **3 possible outcomes**: left pan heavier, right pan heavier, or balanced. Therefore, in n weighings, the total number of distinct outcome sequences is 3^n. This is the "information channel's" capacity to convey different results.
  The problem has **24 initial possibilities** (12 coins × 2 possibilities: heavier or lighter). To uniquely identify one possibility from 24, we need an outcome sequence to be able to represent at least 24 different states. We need the smallest n such that $3^n \geq 24$.
  - $3^2 = 9$ (< 24, not enough)
  - $3^3 = 27$ (≥ 24, sufficient)
    Therefore, a minimum of 3 weighings is theoretically required, and a strategy exists to achieve this limit . The design of the weighing strategy is analogous to constructing a ternary error-correcting code (like a Hamming code in base 3), where each coin and its counterfeit status (heavy/light) is assigned a unique ternary "syndrome" that corresponds to the outcomes of the three weighings